

EXPERIMENT-12: React student management application

Create a React application for the student management system with registration, login, contact, and about pages, and implement routing to navigate through these pages.

Objective : Build a modern React SPA with multiple pages and client-side routing.

Tools : React, React Router, frontend routing, UI components

Outcome : A React-based student management app with page navigation and user registration/login.

Answer:

Build React page components and route them with React Router, and manage registration and login state using local state or context.

File Name : student_app.html

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Student Management System</title>

<!-- React + React Router (CDN) -->
<script crossorigin src="https://unpkg.com/react@18.2.0/umd/react.production.min.js"></script>
<script crossorigin src="https://unpkg.com/react-dom@18.2.0/umd/react-dom.production.min.js"></script>
<script crossorigin src="https://unpkg.com/@remix-run/router@1.14.0/dist/router.umd.min.js"></script>
<script crossorigin src="https://unpkg.com/react-router@6.21.0/dist/umd/react-router.production.min.js"></script>
<script crossorigin src="https://unpkg.com/react-router-dom@6.21.0/dist/umd/react-router-dom.production.min.js"></script>
<script crossorigin src="https://unpkg.com/@babel/standalone@7.23.5/babel.min.js"></script>

<style>
*{box-sizing:border-box}
body{margin:0;font-family:Arial,sans-serif;background:#f4f6f8;color:#222}
nav{background:#2c3e50;padding:12px 24px;display:flex;gap:18px;align-items:center;flex-wrap:wrap}
nav .brand{color:#fff;font-weight:bold;font-size:18px;margin-right:auto}
```

```

nav a{color:#cfd8dc;text-decoration:none;padding:4px 8px;border-radius:4px}
nav a.active{background:#1abc9c;color:#fff}
nav button{background:#e74c3c;color:#fff;border:0;padding:6px 12px;border-
radius:4px;cursor:pointer}
.container{max-width:800px;margin:30px auto;padding:0 15px}
.card{background:#fff;padding:24px;border-radius:8px;box-shadow:0 2px 6px rgba(0,0,0,.1)}
h1,h2{margin-top:0}
label{display:block;margin:10px 0 4px;font-weight:bold;font-size:14px}
input,select,textarea{width:100%;padding:9px;border:1px solid #ccc;border-radius:4px;font-
size:14px}
.btn{margin-top:14px;background:#1abc9c;color:#fff;border:0;padding:10px 18px;border-
radius:4px;cursor:pointer;font-size:15px}
.btn:hover{background:#16a085}
.error{background:#fdecea;color:#c0392b;padding:8px;border-radius:4px;margin-top:10px}
.success{background:#e8f5e9;color:#2e7d32;padding:8px;border-radius:4px;margin-top:10px}
table{width:100%;border-collapse:collapse;margin-top:12px}
th,td{border:1px solid #ddd;padding:8px;text-align:left}
th{background:#2c3e50;color:#fff}
.del{background:#e74c3c;color:#fff;border:0;padding:4px 10px;border-radius:4px;cursor:pointer}
.hero{text-align:center;padding:30px 10px}
</style>
</head>
<body>
<div id="root"></div>

```

```

<script type="text/babel">
const { useState, useContext, createContext } = React;
const { HashRouter, Routes, Route, NavLink, Navigate, useNavigate } = ReactDOM;

// ----- Helper: state saved in localStorage -----
function useLocal(key, initial) {
  const [value, setValue] = useState(() => {
    try { const s = localStorage.getItem(key); return s ? JSON.parse(s) : initial; }
    catch { return initial; }
  });
  const save = v => { setValue(v); localStorage.setItem(key, JSON.stringify(v)); };

```

```

    return [value, save];
  }

// ----- Context: registration + login state -----
const AuthContext = createContext();

function AuthProvider({ children }) {
  const [users, setUsers] = useLocal('students', []);
  const [currentUser, setCurrentUser] = useLocal('currentUser', null);

  const register = data => {
    if (users.some(u => u.email === data.email)) return 'Email already registered';
    setUsers([...users, { id: Date.now(), ...data }]);
    return null; // no error
  };

  const login = (email, password) => {
    const u = users.find(u => u.email === email && u.password === password);
    if (!u) return 'Invalid email or password';
    setCurrentUser({ id: u.id, name: u.name, email: u.email });
    return null;
  };

  const logout = () => setCurrentUser(null);
  const removeStudent = id => setUsers(users.filter(u => u.id !== id));

  return (
    <AuthContext.Provider value={{ users, currentUser, register, login, logout, removeStudent }}>
      {children}
    </AuthContext.Provider>
  );
}

const useAuth = () => useContext(AuthContext);

// ----- Navbar -----
function Navbar() {
  const { currentUser, logout } = useAuth();
  const navigate = useNavigate();

```

```

return (
  <nav>
    <span className="brand">🎓 Student Management</span>
    <NavLink to="/" end>Home</NavLink>
    <NavLink to="/about">About</NavLink>
    <NavLink to="/contact">Contact</NavLink>
    {currentUser ? (
      <>
        <NavLink to="/students">Students</NavLink>
        <span style={{ color: 'fff' }}>Hi, {currentUser.name}</span>
        <button onClick={() => { logout(); navigate('/login'); }}>Logout</button>
      </>
    ) : (
      <>
        <NavLink to="/login">Login</NavLink>
        <NavLink to="/register">Register</NavLink>
      </>
    )}
  </nav>
);
}

```

// ----- Protected route -----

```

function Protected({ children }) {
  const { currentUser } = useAuth();
  return currentUser ? children : <Navigate to="/login" replace />;
}

```

// ----- Pages -----

```

function Home() {
  const { currentUser, users } = useAuth();
  return (
    <div className="card hero">
      <h1>Welcome to the Student Management System</h1>
      <p>Register, log in and manage student records.</p>
      <p><b>Total registered students:</b> {users.length}</p>
    </div>
  );
}

```

```

    {currentUser && <p className="success">You are logged in as {currentUser.email}</p>}
  </div>
);
}

```

```

function About() {
  return (
    <div className="card">
      <h1>About</h1>
      <p>This is a Single Page Application built with <b>React</b> and <b>React Router</b>.</p>
      <ul>
        <li>Client-side routing between pages without reloading</li>
        <li>Registration and login state handled with React Context</li>
        <li>Data saved in the browser's localStorage</li>
        <li>Protected route: the Students page needs login</li>
      </ul>
    </div>
  );
}

```

```

function Contact() {
  const [form, setForm] = useState({ name: "", email: "", message: "" });
  const [sent, setSent] = useState(false);
  const [messages, setMessages] = useLocal('messages', []);
  const change = e => setForm({ ...form, [e.target.name]: e.target.value });
  const submit = e => {
    e.preventDefault();
    setMessages([...messages, { ...form, time: new Date().toLocaleString() }]);
    setForm({ name: "", email: "", message: "" });
    setSent(true);
  };
  return (
    <div className="card">
      <h1>Contact Us</h1>
      <form onSubmit={submit}>
        <label>Name</label>

```

```

        <input name="name" value={form.name} onChange={change} required />
        <label>Email</label>
        <input name="email" type="email" value={form.email} onChange={change} required />
        <label>Message</label>
        <textarea name="message" rows="4" value={form.message} onChange={change} required />
        <button className="btn">Send</button>
    </form>
    {sent && <div className="success">Thank you! Your message was saved.</div>}
</div>
);
}

```

```

function Register() {
  const { register } = useAuth();
  const navigate = useNavigate();
  const [form, setForm] = useState({ name: "", email: "", password: "", course: 'Computer Science' });
  const [error, setError] = useState("");
  const change = e => setForm({ ...form, [e.target.name]: e.target.value });
  const submit = e => {
    e.preventDefault();
    if (form.password.length < 4) return setError('Password must be at least 4 characters');
    const err = register(form);
    if (err) return setError(err);
    alert('Registration successful! Please login.');
```

navigate('/login');

```

  };
  return (
    <div className="card">
      <h1>Student Registration</h1>
      <form onSubmit={submit}>
        <label>Full Name</label>
        <input name="name" value={form.name} onChange={change} required />
        <label>Email</label>
        <input name="email" type="email" value={form.email} onChange={change} required />
        <label>Password</label>

```

```

        <input name="password" type="password" value={form.password} onChange={change}
        required />
        <label>Course</label>
        <select name="course" value={form.course} onChange={change}>
            <option>Computer Science</option>
            <option>Electronics</option>
            <option>Mechanical</option>
            <option>Civil</option>
        </select>
        <button className="btn">Register</button>
    </form>
    {error && <div className="error">{error}</div>}
</div>
);
}

```

```

function Login() {
    const { login } = useAuth();
    const navigate = useNavigate();
    const [form, setForm] = useState({ email: "", password: "" });
    const [error, setError] = useState("");
    const change = e => setForm({ ...form, [e.target.name]: e.target.value });
    const submit = e => {
        e.preventDefault();
        const err = login(form.email, form.password);
        if (err) return setError(err);
        navigate('/students');
    };
    return (
        <div className="card">
            <h1>Login</h1>
            <form onSubmit={submit}>
                <label>Email</label>
                <input name="email" type="email" value={form.email} onChange={change} required />
                <label>Password</label>

```

```

    <input name="password" type="password" value={form.password} onChange={change}
    required />
    <button className="btn">Login</button>
  </form>
  {error && <div className="error">{error}</div>}
</div>
);
}

```

```

function Students() {
  const { users, removeStudent, currentUser } = useAuth();
  const [search, setSearch] = useState("");
  const list = users.filter(u => u.name.toLowerCase().includes(search.toLowerCase()));
  return (
    <div className="card">
      <h1>Student Records</h1>
      <input placeholder="Search by name..." value={search} onChange={e =>
        setSearch(e.target.value)} />
      <table>
        <thead><tr><th>#</th><th>Name</th><th>Email</th><th>Course</th><th>Action</th></tr>
        </thead>
        <tbody>
          {list.length === 0 && <tr><td colspan="5">No students found</td></tr>}
          {list.map((u, i) => (
            <tr key={u.id}>
              <td>{i + 1}</td><td>{u.name}</td><td>{u.email}</td><td>{u.course}</td>
              <td>
                {u.id === currentUser.id
                  ? '(you)'
                  : <button className="del" onClick={() => window.confirm('Delete ' + u.name + '?') &&
                    removeStudent(u.id)}>Delete</button>}
              </td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}

```

```
);  
}
```

```
function NotFound() {  
  return <div className="card"><h1>404 - Page not found</h1></div>;  
}
```

```
// ----- App with routing -----
```

```
function App() {  
  return (  
    <AuthProvider>  
      <HashRouter>  
        <Navbar />  
        <div className="container">  
          <Routes>  
            <Route path="/" element={<Home />} />  
            <Route path="/about" element={<About />} />  
            <Route path="/contact" element={<Contact />} />  
            <Route path="/register" element={<Register />} />  
            <Route path="/login" element={<Login />} />  
            <Route path="/students" element={<Protected><Students /></Protected>} />  
            <Route path="*" element={<NotFound />} />  
          </Routes>  
        </div>  
      </HashRouter>  
    </AuthProvider>  
  );  
}
```

```
ReactDOM.createRoot(document.getElementById('root')).render(<App />);  
</script>  
</body>  
</html>
```

